

General Relativity

Final Examination

Spring 2022

Total Possible Score: 100

Name: _____ Class: _____ Student Number: _____

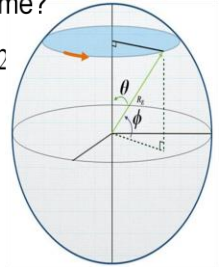
1. (20 points) In Minkowski spacetime, consider a coordinate system S' with coordinates (t', x', y', z') related to the canonical Cartesian coordinate system S , with coordinates (t, x, y, z) , by the transformation

$$t = \left(\frac{c}{\kappa} + \frac{x'}{c} \right) \sinh \left(\frac{\kappa t'}{c} \right) \quad x = \left(x' + \frac{c^2}{\kappa} \right) \cosh \left(\frac{\kappa t'}{c} \right) - \frac{c^2}{\kappa} \quad y = y' \quad z = z'$$

where κ is a (not dimensionless) constant and c is the speed of light.

If an amount of time $\Delta t' = T'$ passes in coordinate system S' , what is the difference in the times recorded by a clock at rest in S' at $x' = 0$ and a clock at rest in S' at $x' = X'_0$? Which clock measures the greater elapsed time?

2. (15 points) Determine whether or not an object moving on the surface of a sphere, i.e., on S^2 around a circle of constant colatitude $\theta = \text{latitude}$ (纬度) ϕ is moving along a geodesic. Note that the value for θ is arbitrary: it need NOT be the value shown in the picture. You may freely use any equations you find in Section 3.7 of the textbook.



3. (20 points) (a) Show explicitly that the commutator of covariant derivatives acting on a scalar function f is zero, that is, $[\nabla_\mu, \nabla_\nu] f = 0$. Is there any extra requirement for this to be true? (b) Given (a), consider scalar functions g and h and a dual vector given by $\omega_\mu = g \nabla_\mu h$.

CLEARLY show that $\omega_{[\mu} \nabla_\nu \omega_{\sigma]} = 0$.

4. (15 points) A two-dimensional space has a line element given by

$$ds^2 = \left(\frac{a}{u \ln u} \right)^2 (du^2 + u^2 dv^2) \quad a = \text{constant}.$$

We are also given one element of the Riemann curvature tensor, $R^u_{vvu} = \left(\frac{1}{\ln u} \right)^2$. If we are also told this space is maximally symmetric, then what is the Ricci scalar equal to?

5. (30 points) The spacetime represented by the line element

$$ds^2 = -dt^2 + dr^2 + (r^2 + \alpha^2)(d\theta^2 + \sin^2 \theta d\phi^2) \quad \alpha = \text{constant},$$

is static, invariant under $t \rightarrow -t$. So any Christoffel symbol with one or more t -index is zero. Christoffel symbols with all indices the same are also all zero.

(a) Of the remaining symbols, nine are non-zero. Find the values for these nine Christoffel symbols.

(b) Write down all the geodesic equations.

(c) Find all Riemann curvature tensor terms of the form R^a_{bab} where $a \neq t$ and $b \neq t$. Find both rank-(1,3) and rank-(0,4) terms.

(d) Calculate the Ricci tensor and the Ricci scalar.

(e) For this spacetime, and in this coordinate system, find the components of the energy-momentum tensor, $T_{\mu\nu}$.

(f) Is there any significance to your answer in (e)?

Hint: consider the "top-down approach" as we recently discussed in lecture.